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PAPER

Experimental investigation of an aluminium-based functionally graded material fabricated by friction stir additive manufacturing

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Abstract

Advanced engineering materials that provide better mechanical performance via small changes in composition and microstructure are called Functionally Graded Materials (FGMs These materials are dominant in the aerospace and automotive industries when property gradients are a crucial factor. This research paper explores Aluminium-based functional graded materials (AFGMs) fabricated utilizing the friction stir additive manufacturing (FSAM). It is designed to determine the influence of tool rotational speed (TRS) on the mechanical properties and the microstructure formation. Layers of increasingly strong alloys AA1100, AA5083, AA6061, AA2024, and AA7075 were produced to form a five-layer AFGM system. The TRS range of 400 to 800 RPM was used for processing. The finest mechanical parameters, including a microhardness of 139 HV, a tensile strength of 310 MPa, a yield strength of 252 MPa, and an elongation of 18%, were obtained by using field emission scanning electron microscopy, which showed that 600 RPM created the finest grain structure. Grain coarsening and reduced mechanical performance were seen at both low and high rotating speeds. To fabricate AFGMs for industrial use using FSAM, these results provide realistic recommendations for parameter selection.

1. Introduction

Functionally Graded Materials are composite systems that exhibit mechanical, thermal, and chemical characteristics that are reliant on their location due to constant spatial changes in composition and microstructure. Due to their compositional gradeds, FGMs can outperform more traditional homogeneous materials under demanding working circumstances. Because of their versatility, FGMs have found their initial aerospace thermal barrier coating use, such as military systems, medical implants, and energy production [1].

Powder metallurgy, thermal spraying, centrifugal casting, and hot pressing are some of the fabrication processes used to make FGM. Each of these methods has its own set of benefits that make it ideal for certain uses. Traditional approaches, on the other hand, often fail to provide the desired results in terms of compositional control, strong interfacial bonding, or homogeneous microstructural distribution. The full potential of FGM cannot be realized without the use of advanced production technologies. Superior control over material distribution, defect reduction, and mechanical property optimization is all possible using friction stir additive manufacturing, a solid-state layer-by-layer deposition approach, for aluminum-based FGM systems [2].

The key to effectively using FGMs is having an adequate understanding of the structure–property correlations. Microstructural transitions that occur gradually increase fracture toughness and reduce stress concentrations at the interface, making the material more resistant to wear and corrosion. FGMs made with FSAM have improved hardness and resilience to cyclic stress because of their refined grain architectures and balanced phase

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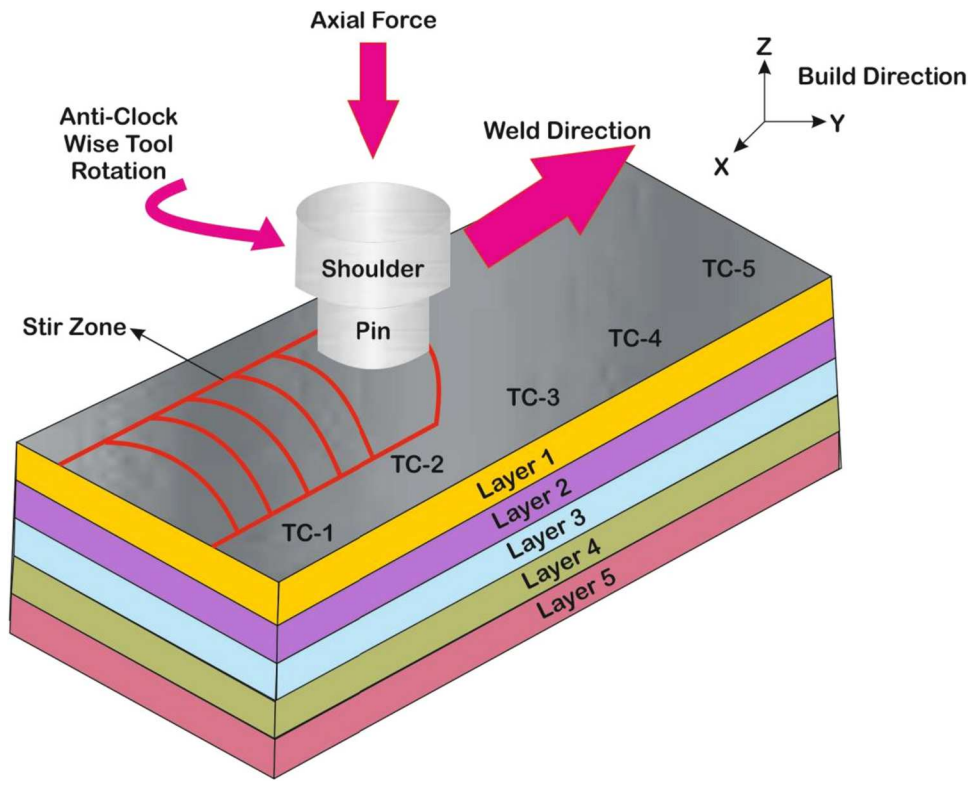


Figure 1. A diagrammatic representation of Friction stir additive manufacturing.

distributions. Therefore, to ensure the maximum FGM performance, it is necessary to choose appropriate fabrication processes, which will give defect-free gradients of compositions that are controlled [3, 4].

Applications of FGMs have increased in a number of industries due to their flexibility and adaptability. Optimal stress distribution, less delamination, and increased thermal resistance are all benefits of using graded compositions as thermal barrier coatings in the aerospace industry. Disc brakes, engine parts, and lightweight structural sections may all benefit from FGMs' wear resistance and thermal management properties [5, 6]. The marine sector uses FGMs for applications that need great mechanical strength and corrosion resistance, such as propeller shafts, turbine blades, and hull components [7, 8]. Due to the precise fabrication control and structural integrity needed for FSAM's effective application in these domains, more research into the technology as a reliable manufacturing process is necessary.

Friction stir additive manufacturing is a solid-state additive manufacturing process that builds upon the principles of friction stir welding, and layer-by-layer deposition through controlled precipitate structures can be designed. Regulating the porosity and purifying the grains, this method produces bespoke FGM gradient topologies while improving mechanical characteristics. A tool in FSAM involves a pin and shoulder rotating tool that makes the material softening but not melting as the tool moves over the substrate to generate frictional heat [9–14]. Under pressure, the plasticized material solidifies, and when it cools, it forms strong metallurgical linkages. This procedure is graphically shown in figure 1, which also shows important FSAM components.

While melting-based approaches may lead to issues including solidifying porosity and hot cracking, FSAM's solid-state nature eliminates these concerns [15]. In contrast, targeted laser gas confinement may cause keyhole porosity in quick melting-solidification cycles [16, 17]. FSAM can suppress temperature gradient, residual strains, and porosity, owing to its ability to avoid melting [10, 18]. Internal cracking remains an issue despite the potential use of electron beam melting to alleviate certain thermal stress issues, especially in materials with low thermal conductivity [19]. Bypassing the detrimental effects of rapid solidification, FSAM has the potential to produce homogeneous grain structures, as opposed to the anisotropic features often seen in selective laser melting [20]. Owing to this variation, FSAM is perceived as an effective way of fabricating aluminum FGMs having controlled gradient designs and enhanced structural strength to be utilized in significant environments.

To ensure the presence of microstructural homogeneity and uniform mechanical behaviour, FSAM addresses the major manufacturing challenges, which include porosity, failure of interlayer bonding, and residual stress. All these characteristics bring FSAM a promising alternative to the conventional additive manufacturing technologies that rely on fusion. Process factors influence the heat production, the flow of materials, and the

attributes that develop, including the velocity at which the tool is rotated. Comprehending how TRS affects FGM production quality and performance is crucial to process optimization. This study addresses this gap by conducting a systematic investigation of the impacts of TRS on the mechanical properties and layer-by-layer formation of the microstructure of AFGMs produced by FSAM. Besides, this research examines the reliability of the overall process, the consistency of interfaces among materials, and the development of the microstructure of AFGMs in extreme conditions.

The primary objectives of this research are as follows:

(i) To systematically characterize the microstructural features across multiple layers and zones of the AFGM fabricated at different TRS.

(ii) To quantify and analyze the elemental composition and its variation across five distinct layers under varying TRS conditions.

(iii) To investigate the relationship between microstructural evolution and corresponding mechanical properties (tensile strength, hardness) at various TRS.

(iv) To determine the optimal TRS that provides the best balance between microstructural refinement and mechanical performance of AFGM.

The uniqueness of this paper is the fabrication and systematic assessment of a five-layer aluminium-based functional gradient material of AA7075, AA2024, AA6061, AA5083, and AA1100 built in a strength-graded arrangement. This multi-alloy structure is a contrast to the binary or two-layered systems studied in the past through FSAM, where complex metallurgical gradients are created, which facilitates more stringent evaluation of layer-wise recrystallization and diffusion of elements. These advances in the study, as it explores a large TRS (400–800RPM) range and develops a clear process structure property relationship between heat input, grain refinement, precipitation behavior, hardness distribution, and tensile response. The result shows that improved microstructures and high mechanical performance can be achieved with optimized TRS, and no reinforcing additions are required, overcoming the drawbacks of previous studies on FSAM-based FGM.

2. Related works

FSAM has gained significant attention as a solid state route for producing AFGMs with minimized solidification defects and controlled microstructure evolution [1–3], unlike fusion based additive processes, FSAM enables multilayer deposition through thermo-mechanical stirring, promoting refined equiaxed grains, stable precipitate structures and superior bonding quality prior research on AFGMs fabricated through friction stir based techniques have primarily focused on binary or ternary alloys system, or reinforcement assisted gradients using ceramic or nano-particle additions, often achieving improvements in hardness and wear resistance but with limited control over through-thickness microstructural uniformity [4–8]. Studies that attempted multi-layer deposition using more than two alloys typically reported issues such as interfacial inconsistency, heat accumulation, and significant variability in tensile performance at higher process temperature [9, 10, 21]. Thus, literature addressing complex, multi-alloy, reinforcement-free aluminium FGMs fabricated entirely by FSAM remains sparse.

Addressing these gaps, the present study investigates a five-layer aluminium FGM comprising AA7075, AA2024, AA6061, AA5083, and AA1100 arranged according to their strength contribution. The work systematically evaluates the effect of TRS between 400 and 800RPM as microstructural evolution, elemental diffusion, defect generation, and mechanical behaviour, thereby establishing a comprehensive process structure property framework for multi-alloy FSAM systems.

TRS is consistently identified as the dominant FSAM parameters governing thermal input, material plasticization and stirring efficiency [1, 3, 11] several investigations across friction stir based additive processes have shown that insufficient TRS leads to inadequate heat generation and poor layer consolidations, while excessive TRS promotes grain coarsening, precipitate over-aging, interlayer porosity and microcrack formation due to thermal overshoot [11–14]. Optimal microstructural refinement is usually achieved within a mid-range TRS window, yet most available studies have examined single alloys [AA6061, AA7075] or simple bimetallic stacks where thermal gradient and solute diffusion behaviors are relatively predictable [6, 12, 21]. For multi-alloy FSAM, especially those involving alloys of different thermal conductivity and precipitation kinetics, selecting a moderate TRS range (e.g. 400–800RPM) is crucial to maintain sufficient heat input while avoiding thermal degradation. Despite this, comprehensive TRS optimization studies on multi-alloy aluminium FGMs remain highly limited.

Table 1. Chemical compositions of the Aluminium alloys (base material).

Layers	Alloys	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Remaining
Layer -1	AA7075	—	—	1	—	1.8	0.1	4.8	—	Al
Layer -2	AA2024	0.49	3.9	0.4	1.4	0.8	0.2	0.2	0.1	Al
Layer -3	AA6061	0.6	0.6	0.2	0.15	0.9	0.1	0.2	0.1	Al
Layer -4	AA5083	0.35	0.3	0.7	0.4	4	0.05	0.2	0.1	Al
Layer -5	AA1100	0.4	0.3	0.09	—	—	—	—	—	Al

Layer architecture is another critical factor influencing mechanical performance and functional response. Existing works on AFGMs typically generate gradients by modifying surface layers, adjusting reinforcement fractions, or stacking alloys with minimal strength differentials [4, 7, 9]. However, the systematic arrangement of wrought aluminium alloys based on strength hierarchy and metallurgical compatibility, such as integrating high-strength, precipitation hardenable alloys (AA7075, AA2024), medium-strength precipitation alloys (AA6061), strain-hardened solid solution alloys (AA5083), and ductile base layers (AA1100), has received only limited attention. Such an arrangement aligns with the fundamental functionally gradient principle, where properties transition smoothly through the thickness according to the local mechanical requirement, ensuring that high-strength materials are placed in load-bearing regions while more ductile alloys mitigate strain concentrations and enhance overall structural stability. Strength-based stacking also adheres to the FGM rule of gradual property variation, which states that the FGM should exhibit continuous or step-wise transition in composition, microstructure, and mechanical properties to reduce residual stresses, suppress interfacial weakening, and improve load transfer efficiency. Despite its theoretical advantages, only a few studies have experimentally validated this design philosophy using FSAM, and a comprehensive assessment of how such gradient architectures perform. The influence of interlayer bonding, in such architectures, on diffusion behavior and long-term mechanical integrity remains scarce.

From this review, several critical research gaps are evident:

(i) A lack of systematic studies on multi-alloy, reinforcement-free aluminium FGMs fabricated via FSAM, particularly those involving five or more dissimilar alloys.

(ii) Insufficient understanding of how TRS within the 400–800RPM window influences grain refinement, precipitate evolution, and defect formation in complex multi-layer stacks.

(iii) Limited quantitative correlation between layer-wise microstructure (grain size, precipitate distribution, and compositional gradient) and mechanical response, such as hardness, tensile strength, and Hall-Petch behavior in FSAM fabricated FGMs.

(iv) Scarcity of studies that rationalize alloy-layer arrangement based on mechanical hierarchy and validate resulting process-structure property relationships experimentally.

3. Materials and methods

This study proves that FSAM can be used to fabricate AFGMs. The tensile strengths of five different aluminum alloys were taken into consideration while making the selection: AA1100 (129 MPa), AA5083 (309 MPa), AA6061 (331.56 MPa), AA2024 (465 MPa), and AA7075 (565 MPa). The chemical components are shown in table 1. These alloys find widespread use in maritime and aeronautical constructions, but when used in isolated components, they create production issues that make repairs and maintenance more difficult. The alloys were ordered in descending order of strength to tackle these challenges, as shown in figure 2. The weakest alloy, AA1100, was placed at the bottom, and the strongest, AA7075, was placed at the top. Each layer was prepared with dimensions of 300 mm x 100 mm x 3 mm. The FSAM process was performed using the CNC Friction stir welding machine (Model 2014), equipped with a 3HP motor with brake, a 22 kW spindle capable of achieving up to 5000RPM, and a 20 KN load cell to ensure precise fabrication of the AFGM.

The parameter process was chosen and closely regulated to obtain a perfect bonding between the layers to form a defect-free process. These important parameters, summarized in table 2, entail a tool rotation speed of 400–800 RPM, 20 mm min^{-1} weld Speed, a plunge depth of 14.5 mm, and a tool penetration of 3 mm. These parameters were carefully manipulated to facilitate normal stirring and proper heat generation to facilitate a good flow of materials and good bonding of the solid state in all the layers. Besides, the geometrical layout of the tool, reported in table 3 and figure 3, was specifically created to promote high efficiency in material mixing, as well as facilitate the process of depositing layers in the build process.

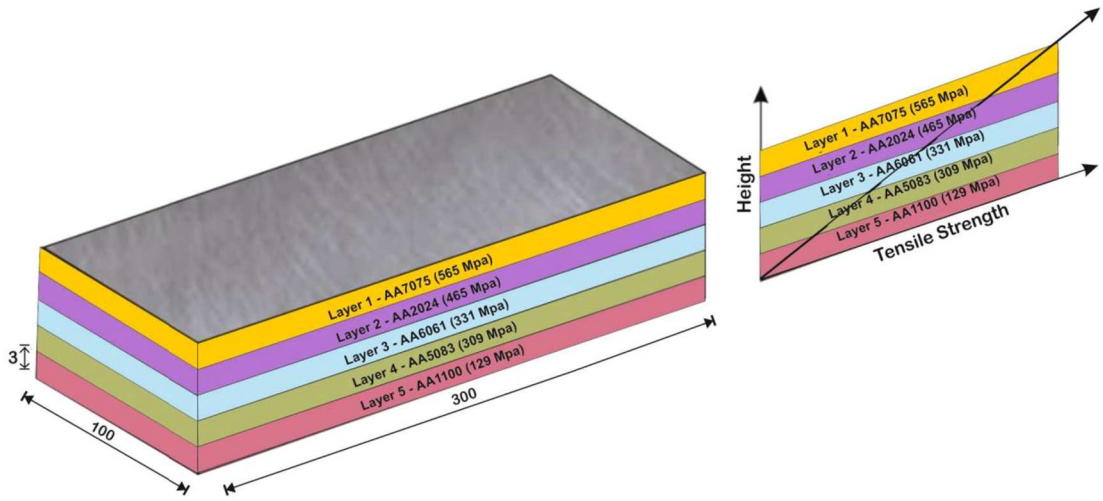


Figure 2. A schematic layer-wise arrangement of the Aluminium alloys.

Table 2. Process parameters used for FSAM to develop AFGM.

Parameters	Values
Tool rotational speed in RPM	400, 500, 600, 700, 800
Weld speed	20 mm min^{-1}
Tool tilt angle	2°
Tool plug rate	3 mm min^{-1}
Number of passes	Single

Table 3. Geometry of tapper-threaded pin-shaped tool.

	Shank	Body	Shoulder	Taper-threaded profile pin	
Diameter (mm)	25	31	30	Top	10
				Bottom	8
Length (mm)	65	40	10	14	

To ensure stable material flow and reproducibility of the FSAM process, all critical process parameters were precisely controlled during fabrication. An axial force of 7–8 kN that is continuous was applied, depending on the selected TRS, to maintain consistent tool-workpiece engagement The rate of plunge was set at 3 mm min^{-1}, followed by a controlled dwell time of 8–10 s before tool traverse to promote adequate frictional heat generation and plasticization. Temperature evolution during FSAM was monitored using K-type thermocouples positioned near the tool path and at key interlayer interfaces to capture the thermal gradient across the multi-alloy stack. A schematic of the experimental setup, including thermocouple locations, tool rotational direction, and traverse direction, has been included (figure 1) to provide a clear representation of the thermal and mechanical conditions governing the multi-layer FSAM process.

The fabricated AFGM was deprived of microstructural, tensile as well as microhardness characterization of specimens. In order to examine the microstructural features of the weld, three major areas of the weld underwent the microstructural analysis as given in figure 4: FGM layering includes advancing side (AS), stir zone (SZ), retreating side (RS), and interlayers. The samples were subjected to a conventional preparation in terms of metallography and this entailed polishing sequentially in order to achieve a mirror finish. The Keller reagent was used to do the etching (Composition: 2.5 ml of hydrofluoric acid, 2.5 ml of hydrochloric acid, and 50 ml of distilled water), and the dwell time was 10 seconds to reveal the morphology of the grain. In order to examine the layer interfaces and finer grain structure, optical microscopes with a power of 100x were used for microstructural investigations.

To have a thorough insight into the microstructural development, FESEM (CARL ZEISS SIGMA 300 FESEM EDAX) was examined on the AS, SZ, and RS in all five layers of the AFGM. FESEM operated at an

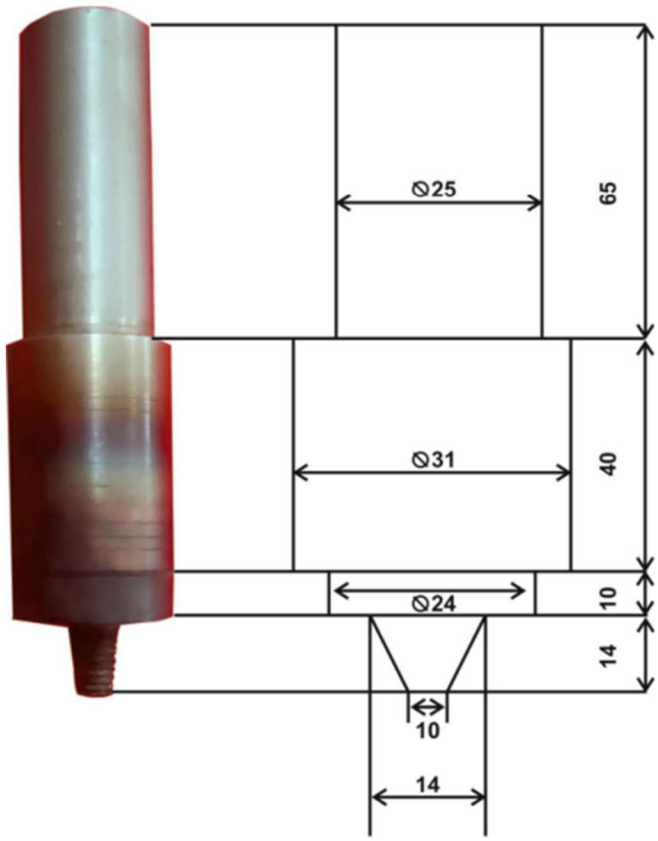


Figure 3. Schematic illustration of the FSAM tool showing the shoulder diameter, pin profile, pin length, and tilt angle used for fabricating the AFGM.

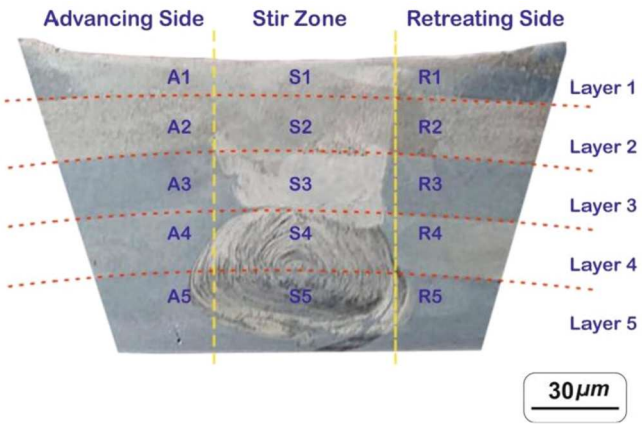


Figure 4. Comprehensive layer-wide picture of AS, SZ, and RS zones.

accelerating voltage of 15 kV, which permitted the high-resolution fine microstructural features, such as grain size, creation of precipitates, and dispersion of the precipitates. These measurements were related to TRS and heat production to measure their effect on refining grains and the behaviour of the precipitates. To measure the distribution of elements and to confirm the compositional consistency throughout the layers confirming the chemical integrity of the process, Complementary Energy-Dispersive X-ray spectroscopy (EDX) analysis was performed.

Tensile specimens were prepared from the constructed AFGM according to ASTM E8/E8M-11 aluminum alloy requirements. Each specimen was accurately machined to meet the tensile testing setup dimensions and alignment standards. Electro-hydraulic Servo Testing Machine WAW-C Series tensile tests were performed. The specimen geometry is shown in figure 5.

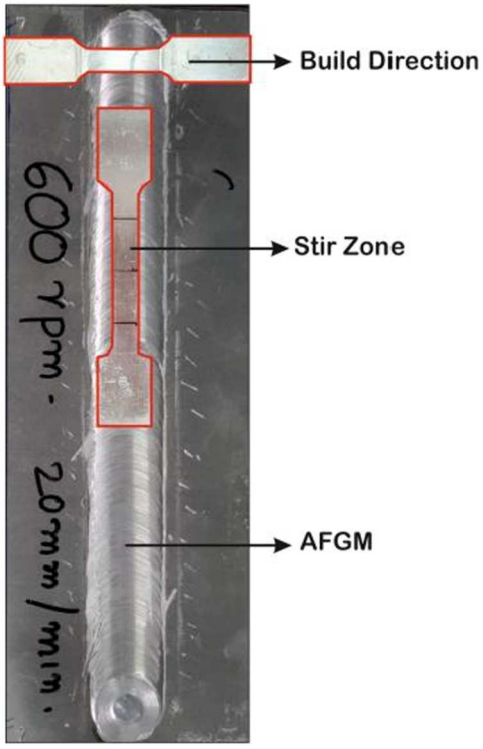


Figure 5. Schematic of FSAM-fabricated AFGM SZ tensile specimen extraction along build direction.

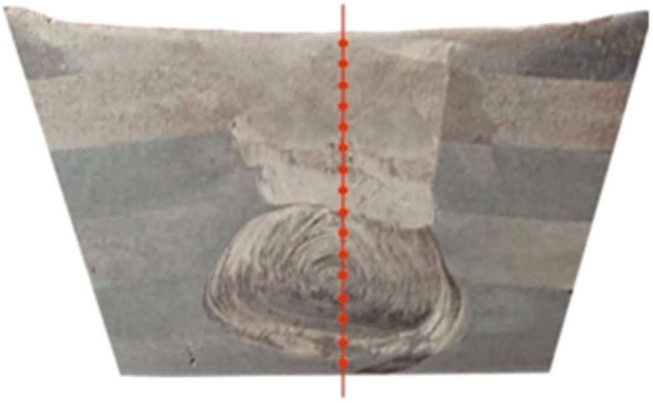


Figure 6. Microhardness profiles across the SZ of AFGM fabricated by FSAM.

A universal testing machine was used to assess the overall strength of the AFGM under assessment to identify ultimate tensile strength (UTS), yield strength, and the percentage of elongation at fracture of every specimen. Microhardness was measured on the cross-section of the top-to-bottom layer using a Vickers hardness tester (Insize HDT-VH). After applying a load of 9.81 N over 10 seconds, indentations were created at regular intervals as shown in figure 6, which gave a profile of hardness throughout the AFGM.

For statistical consistency, all mechanical and microstructural measurements were performed on three specimens extracted from the AFGM. The reported grain size, hardness, and tensile strength values represent the average of these three measurements, ensuring that the results reflect the true behavior of the material in that region. Minor variations observed between samples were accounted for by averaging the readings, and the final values presented in the results section correspond to these averaged values. In addition, repeated measurements were taken at each region to minimize local heterogeneity effects, thereby improving the reliability and reproducibility of the reported results. Wherever applicable, 95% confidence intervals were also considered to ensure the reliability and reproducibility of the reported results.

Table 4. Heat generation during the FSAM.

Tool rotational speed in RPM	Axial Force (F) in KN	Heat generation (Q) in W
400	8	0.51
500	7.5	0.60
600	7	0.67
700	6.5	0.73
800	6	0.77

4. Result

Friction of the revolving tool and the base material is the main source of heat production in FSAM. The amount of heat generated is measured in equation (1). The resultant figures are indicated in the table 4 [22]. When TRS is low (400–500 RPM), the amount of heat generated by friction is low, and therefore, the plasticization is not enough, where the interlayer diffusion does not occur. A TRS of 600 RPM, on the other hand, offered the best heat generation so that refinement of the grains can be done without the grains being degraded by heat. Nevertheless, excessive heat production can be expected at higher TRS (700–800 RPM), which leads to the coarsening and degradation of the grains, which is manifested by the decrease in tensile strength and ductility.

$$Q = \mu F\omega\left(\frac{2}{3} \quad R_{shoulder}{}^{3} - \frac{1}{3}R_{pin}{}^{3}\right. \tag{1}$$

Where,

Q – Heat generation in J/mm
ω = Angular velocity
μ—Coefficient of friction = 0.7
F—Axial force in KN
$R_{shoulder}$—Radius of the shoulder = 15 mm
R_{pin}—Radius of the pin = 5 mm

4.1. Microstructural evaluation

Characterization of the microstructure was carried out in the AS, SZ, and RS areas. Representative micrographs for the optimal TRS of 600 RPM are shown in figures 7–9. All the grain size values reported in this section represent the average of three measurements taken from equivalent locations across each layer, and the associated standard deviations were considered to ensure the reliability of the microstructural trends observed. The detailed layer-by-layer microstructural evaluations for other TRS (400,500,700, and 800RPM) are provided in the figures 18–28 (supplementary file -1). The corresponding layer-wise average grain sizes (in microns) for each TRS are summarized in table 5.

4.1.1. Advancing side (AS)

The AS exhibits an apparent gradient in grain size between layers, which is extremely dependent on TRS. The bottom layer exhibits coarse grains of about 15 μm at 400 RPM, while the top layer exhibits coarse grains of about 12 μm. This indicates limited plastic flow and limited heat input for full dynamic recrystallisation. Grain sizes are reduced to about 10 μm in the top layers and 13–14 μm in the bottom layers at 500 RPM, showing improved but still inadequate stirring and heat conditions.

All layers demonstrate significant grain refinement at 600 RPM (figure 7). Grain sizes vary between 6 to 9 μm in the bottom layers to 5 to 6 μm in the top layer, demonstrating a uniform distribution of heat along with efficient plastic flow. Grain sizes increase to 8μm in the top layers and about 12 μm in the bottom layer at 700 RPM, when the grain growth is visible. Grain sizes range from 15 to 17 μm across layers, and there is noticeable grain coarsening at 800 RPM. Grain growth and dynamic recrystallisation are enhanced by high heat input, as evidenced by the progressive enlargement of the grains at 700–800 RPM.

4.1.2. Stir zone

Grain sizes at 400 RPM varied by layer; the upper parts had considerably smaller grains of roughly 10 μm, while the bottom layers had bigger grains of 12–15 μm. This discrepancy resulted from the top regions' greater ability to evacuate heat. At 500 RPM, the grains grew somewhat smaller, with bottom diameters of 14–16 μm and top diameters of roughly 13 μm. Even while there was still not enough heat or material flow to ensure equal grain dispersion, this meant that the conditions were better than they had been

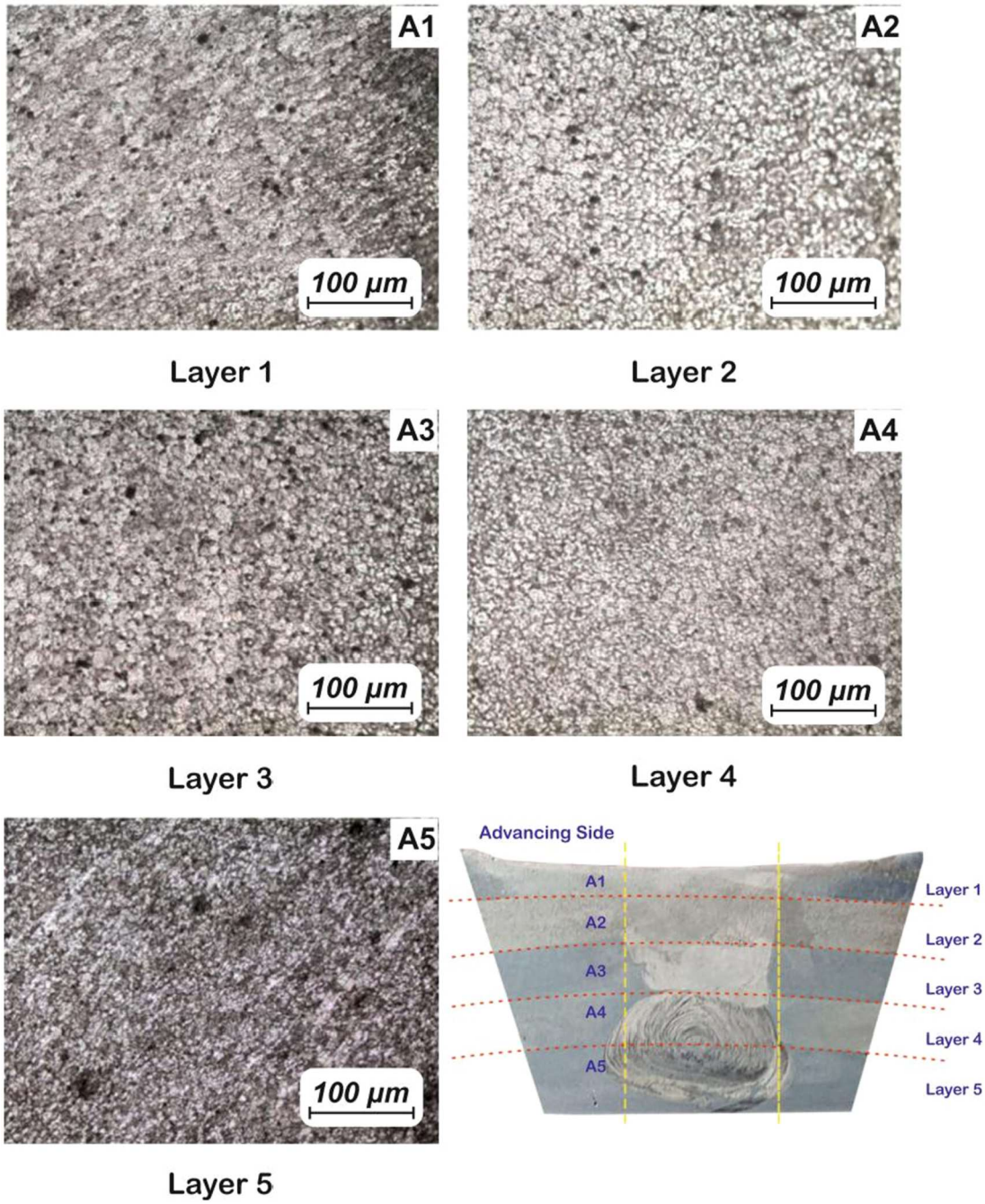


Figure 7. Layer- by Layer-by-layer microstructural evaluation of the AS: AFGM at 600 rpm.

Significant particle size reduction occurred at 600 RPM, where the best results were observed (figure 8). Although the grains of the top layer dropped down to 3 μm in diameter, the grains of the bottom layer dropped down to 7 μm. The obtained results revealed that the right temperature and mechanical balance of homogenous grain refinement between layers were attained at 600 RPM. The grain size began to slightly resume its rise at 700 RPM due to the heat generation. The grain sizes were 9–11 μm at the bottom to 7–8 μm at the top. Grain expansion began to happen as the recrystallization process proceeded efficiently. Lastly, issues were produced by excessive heat input at 800 RPM. Grain expansion was finally encouraged by the prolonged high temperatures, even though the severe deformation and recrystallization initially produced fine grains in the stir zone. As a result, the entire structure saw significant grain coarsening, with diameters reaching 20 μm in the top sections and 21 μm in the bottom parts.

4.1.3. Retreating side

The RS, which undergoes less plastic deformation than the SZ, always has a relatively coarser grain structure. For instance, at 400 RPM, the bottom layers exhibited grain sizes of ~19 μm, while the top layers exhibited finer

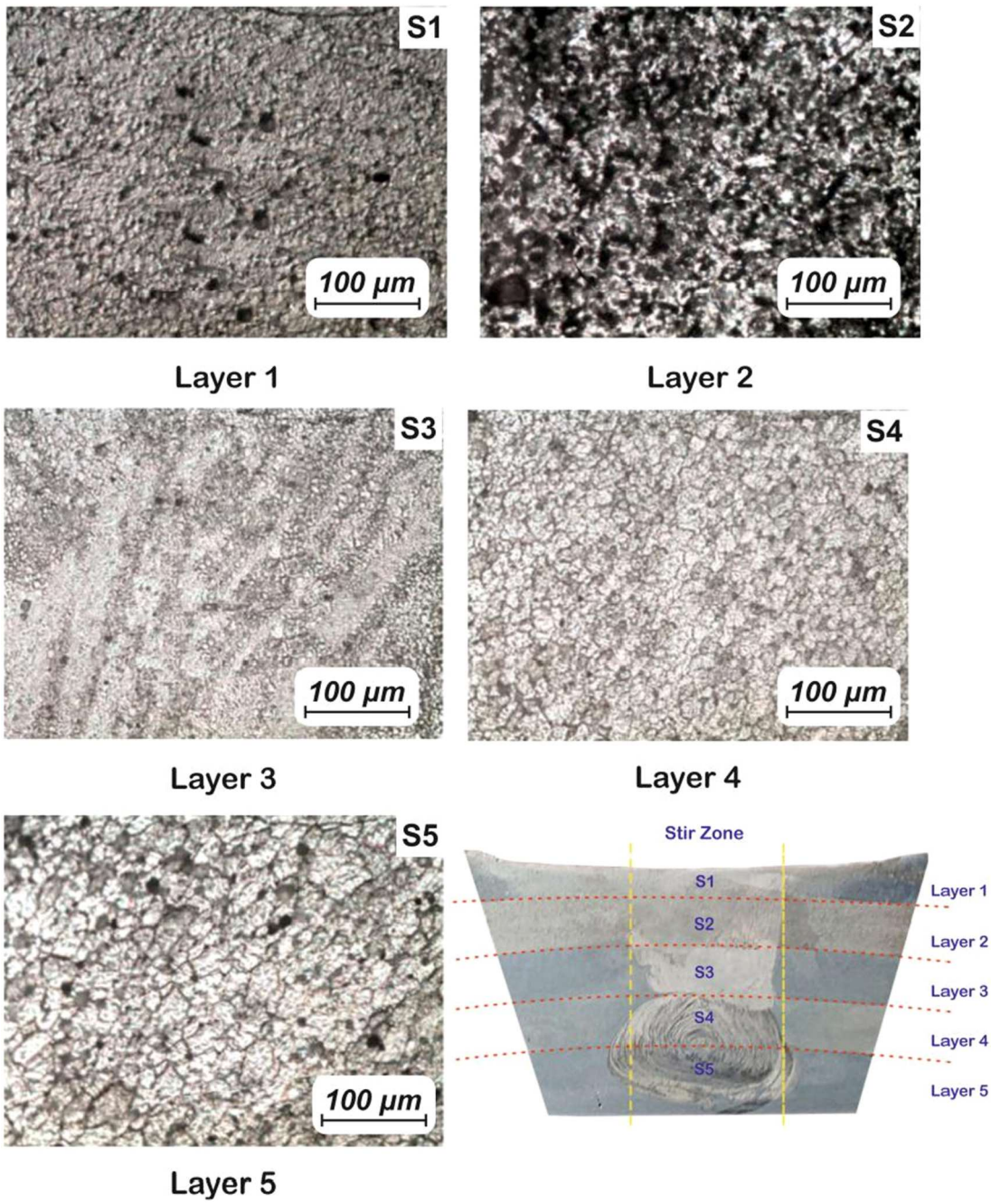


Figure 8. Microstructural study of 600 RPM-graded SZ aluminum-based gradient material.

grains of 15 μm; again, this indicated partial grain refinement as heat dissipated to the surface. At 500 RPM, there was some improvement with the bottom layer sizes reducing to 17–18 μm and 15–16 μm in the top layers, indicating a slightly better heat input and an improved material flow over that at 400 RPM, but grain refinement was still incomplete. At 600 RPM (figure 9), grain sizes were refined to 6 μm in the top layers and 9 μm in the lower layers, indicating optimal heat distribution and efficient material stirring. However, at 700RPM, grain coarsening begins to reemerge, with grain sizes increasing to 11–13 μm in the lower layers and around 10μm in the top layers, suggesting the onset of excessive heat input suppressing the refinement effect. At 800 RPM, sustained high heat caused pronounced grain coarsening, with grain sizes reaching approximately 21 μm in the bottom layers and around 18 μm in the top layer, even in this relatively less-deformed region.

Across all regions, the layer-by-layer observations clearly show that 600 RPM delivers the finest and most consistent grain structure. This rate of dynamic recrystallization prevents grains from overgrowing because the heating energy is exactly suitable for plastic deformation. Even at the lower end, such as 400 RPM, the grains remain relatively coarse in each of the layers, particularly on the AS and RS, because the heat and deformation are not intense enough to initiate complete refinement. Conversely, the system overheats at 800 RPM, and the

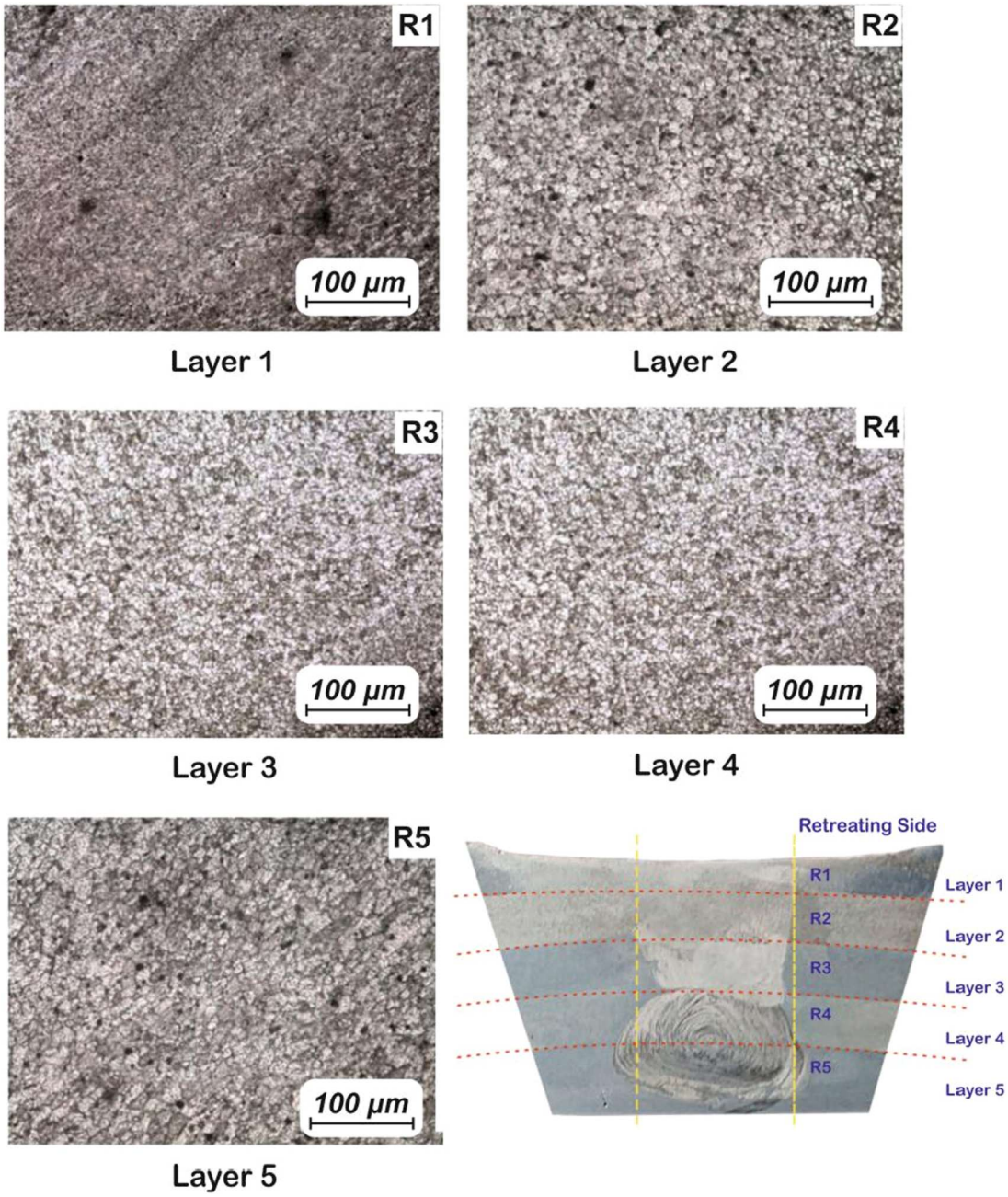


Figure 9. Microstructural evaluation of RS - Aluminium-based functionally gradient material at 600 rpm.

Table 5. Layer-wise Average Grain sizes in microns (µm) according to tool rotational speed.

	400 RPM			500 RPM			600 RPM			700 RPM			800 RPM		
Layers	AS	SZ	RS	AS	SZ	RS	AS	SZ	RS	AS	SZ	RS	AS	SZ	RS
1	12	10	15	10	13	15	5	3	6	8	7	10	15	20	18
2	12	10	15	12	13	16	6	4	7	9	8	10	15	20	17
3	13	12	16	13	14	17	7	4	8	10	9	11	16	20	18
4	14	13	17	14	14	17	6	5	9	11	10	12	16	22	19
5	15	15	19	14	15	18	9	7	9	12	11	13	17	21	21

excess heat leads to a noticeable coarsening of the grains across the layers and zones, which may lead to the ultimate failure of the mechanical performance.

The variation in grain size across the advancing side (AS), stir zone (SZ), and retreating side (RS) reflects the intrinsic asymmetry in strain rate, material flow and thermal history during the FSAM process. The SZ

consistently exhibits the finest grain structure because it experiences the maximum plastic deformation, highest strain rate, and most intense dynamic recrystallization, driven by direct tool-material interaction and severe material stirring. In contrast, the AS shows moderately refined grains, as the tool rotation and traverse directions act in the same sense, producing high shear deformation and effective recrystallization, but with slightly lower strain intensity than the SZ.

The RS, however, displays the coarsest grain structure because the tool rotation opposes the traverse direction, resulting in reduced plastic flow, lower shear strain, and weaker recrystallization compared to both the SZ and AS. Furthermore, the RS tends to retain higher localized heat for a longer duration due to slower material evacuation and reduced convective heat dissipation, which enhances post-recrystallization grain growth. Consequently, the combined effect of maximum deformation in the SZ, intermediated deformation in the AS, and minimum deformation with higher thermal exposure in the RS leads to the observed grain size trend of SZ<AS<RS across all TRS conditions.

4.2. FESEM analysis

The variation in the microstructure of all five layers during the advancing side, stir zone, and retreating side of the tool with varying tool rotation speeds was observed using field emission scanning electron microscopy. Some of the main properties were examined, such as the size of the grains, their distribution, and the presence or absence of pitting and stability of the overall microstructure. It was evident that there were differences between samples that were processed at different rpms of 400 and 800. Figures 10–12 indicate the micrographs of the samples, at the most uniform and refined microstructure was produced at 600 RPM. To reduce it, other rotation speed micrographs are provided in the supplementary file 2 (figures 29–40) to allow those who desire to make detailed comparisons.

4.2.1. Advancing side (AS)

The tool's rotational speed significantly influenced precipitate formation and distribution across layers. At 400 RPM, insufficient heat generation and limited material flow caused extensive precipitate clustering, particularly of magnesium and zinc-rich phases. Layer 3 exhibited magnesium concentrations reaching 2.7%, adversely affecting material ductility and strength. Improving structural regularity and mechanical qualities inside the advancing side, increasing TRS to 500 RPM, and increasing precipitate dispersion. Layers 2, 3, and 4 developed a refined microstructure with improved corrosion resistance when processed at 600 RPM (figure 10), thanks to the fine, evenly dispersed precipitates. The coarsening of precipitates and the preferential concentration of phases rich in magnesium and copper in the outer layers (1 and 5) were the hallmarks of the microstructural deterioration that began at 700 RPM. The ductility and tensile strength were severely reduced at 800 RPM due to the coarse precipitate morphologies caused by the extreme clustering of magnesium, zinc, and copper-bearing precipitates inside these peripheral layers, caused by high heat input.

4.2.2. Stir zone

The TRS was closely related to the refinement of the precipitate in the SZ. The precipitates were also large and irregularly distributed at 400 RPM, especially in layer 3, where the diffusion of Mg was less because of low heat generation. The precipitates were better dispersed after increasing the TRS speed to 500 RPM; the structure and mechanical reaction were more homogeneous, and the Mg and Cu particles were more uniformly distributed in layers 2 and 3. Figure 11 shows that the refinement was best at 600 RPM, which is where the particles of Mg and silicon (Si) were uniformly distributed throughout all layers, generating the highest strength and deformation resistance. At 700 RPM, some coarsening was seen in the fifth layer, because the higher speed produced the larger particle size of Mg and Cu precipitates. The overall precipitation development and aggregation, as well as the compromised mechanical integrity of the layers 1–5, were facilitated by the higher velocity at 800 RPM as the elements had a very extensive distribution.

4.2.3. Retreating side

Clusters of Mg and Zn were most frequently found in layer 3 and had very low refinement, as there was not enough heat input. Moderate improvement was observed at 500 RPM, and the particles of Mg and Cu were more evenly distributed in layers 2 and 3. At 600RPM, the grains (figure 12) were well distributed throughout all the layers and enhanced the mechanical properties of the RS. Precipitate coarsening was observed at 700 RPM in layers 1, 4, besides 5, so that ductility was impaired since bigger Mg precipitates were formed. Precipitate was observed in layer 1 and layer 5 at 800 RPM, and the clusters of coarse Mg, Cu, and Zn layers weaken the structural integrity this leading to low strength and low resistance to corrosion because of poor homogeneity.

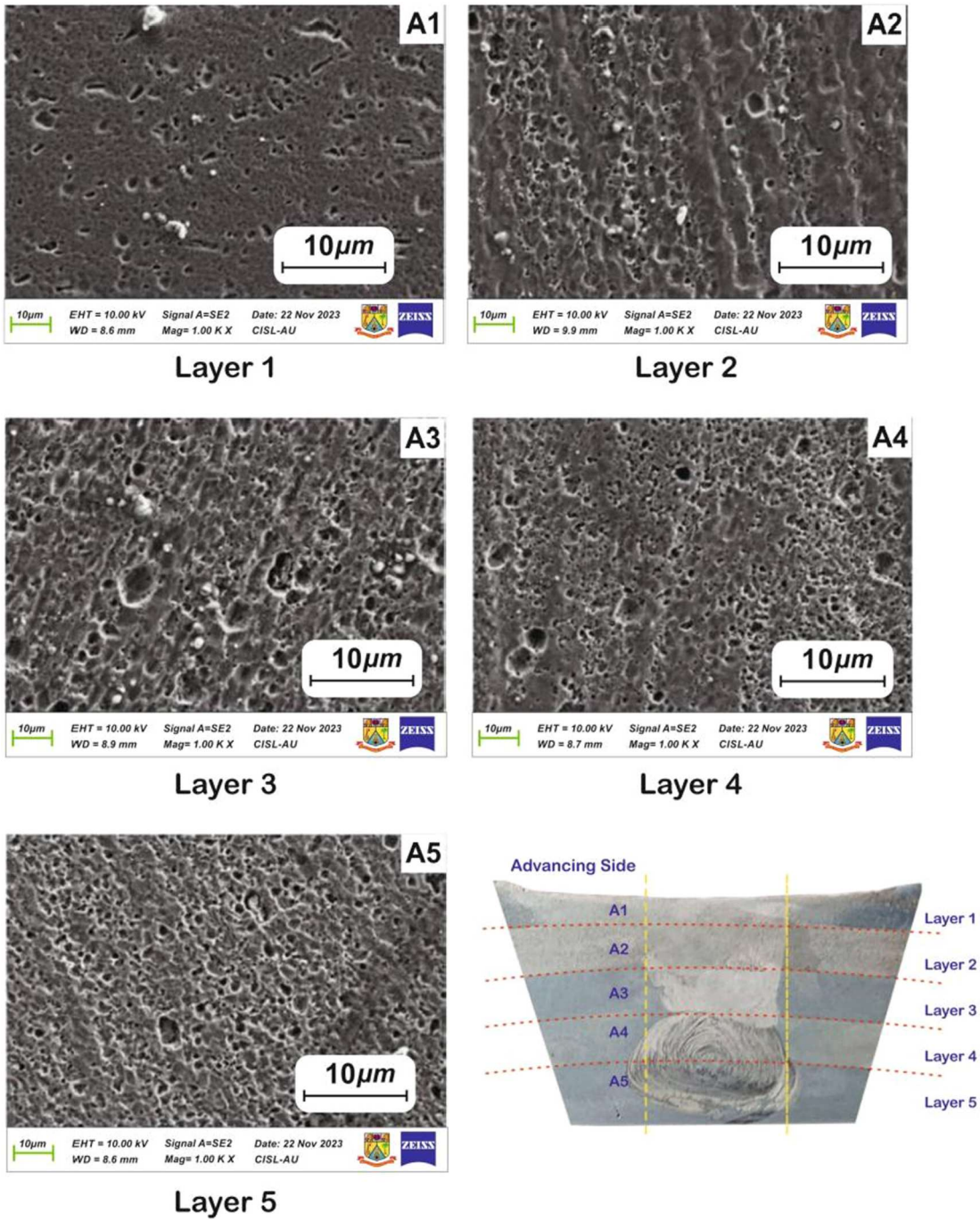


Figure 10. Layer-by-layer microstructural representation of AS using FESEM - aluminium-based functionally gradient material developed by FSAM.

4.3. Chemical composition

EDX was used to describe the elemental distribution across the AS, RS, and SZ of the FSAM samples. Clear chemical differences were noted between these zones, which could be explained by the action of TRS and the material flow dynamics. All the elemental distributions of alloying elements, at the layer-by-layer level, are shown in figures 41 through 55 (supplementary file - 3), and confirm the systematic differences in distribution of alloying elements.

4.3.1. Advancing side

The concentration of Mg in the top layer is comparatively stable (2.5–2.7), but the concentration of Mg in the fifth layer is broader (0.9–4.5), which means that the elements become unevenly incorporated, which is probably due to the decrease in the rate of movement of the material at a relatively deep level. The gradient with Si is a comparable one, with 0.6–0.8 percent in the upper layer, while the bottom one contains 0.2–0.4 percent.

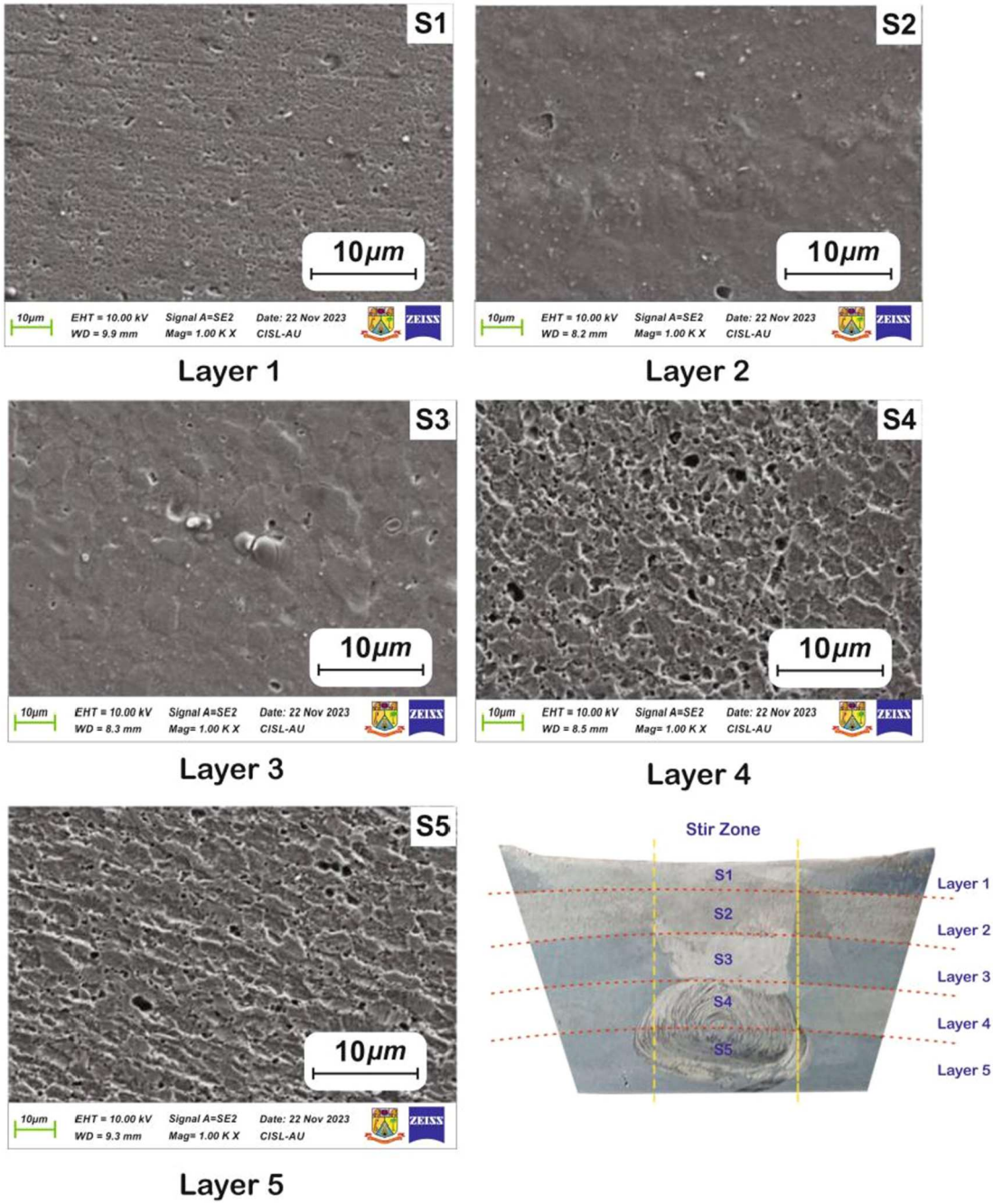


Figure 11. Layer-by-layer microstructural depiction of SZ utilizing FESEM—FSAM's aluminum-based functional gradient material.

The top contains high concentrations of Cu and Zn, with the former ranging between 1.5–2.5, and the latter between 5.5–6, with a sharp decline in 5th layer to about 0.1–0.2 and 0.1 percent, respectively. Such a progressive decrease from top to bottom implies that in high TRS, heavier elements are more likely to be concentrated near the surface with little penetration into deeper layers.

4.3.2. Stir zone

In the Stir zone (SZ), the elemental distribution—much more uniform on the layers than one would expect from the material's basic components (figure 13). The content of magnesium, for example, is quite constant, declining only slightly to 2%–3.8% even at the bottom five layers, and at the top of or within a range of 2.5%–2.7%. Although not as much as the advancing or retreating sides, silicon, copper, and zinc do taper with depth. Silicon starts at 0.7%–0.8% at the surface and decreases to 0.3–0.4 at the fifth layer. Zinc is reduced to 5.5–6 on the surface to about 0.1 on the bottom, while copper is reduced to 1.5–2.3 to a negligible 0.1–0.2—again, the tendency is more gradual here.

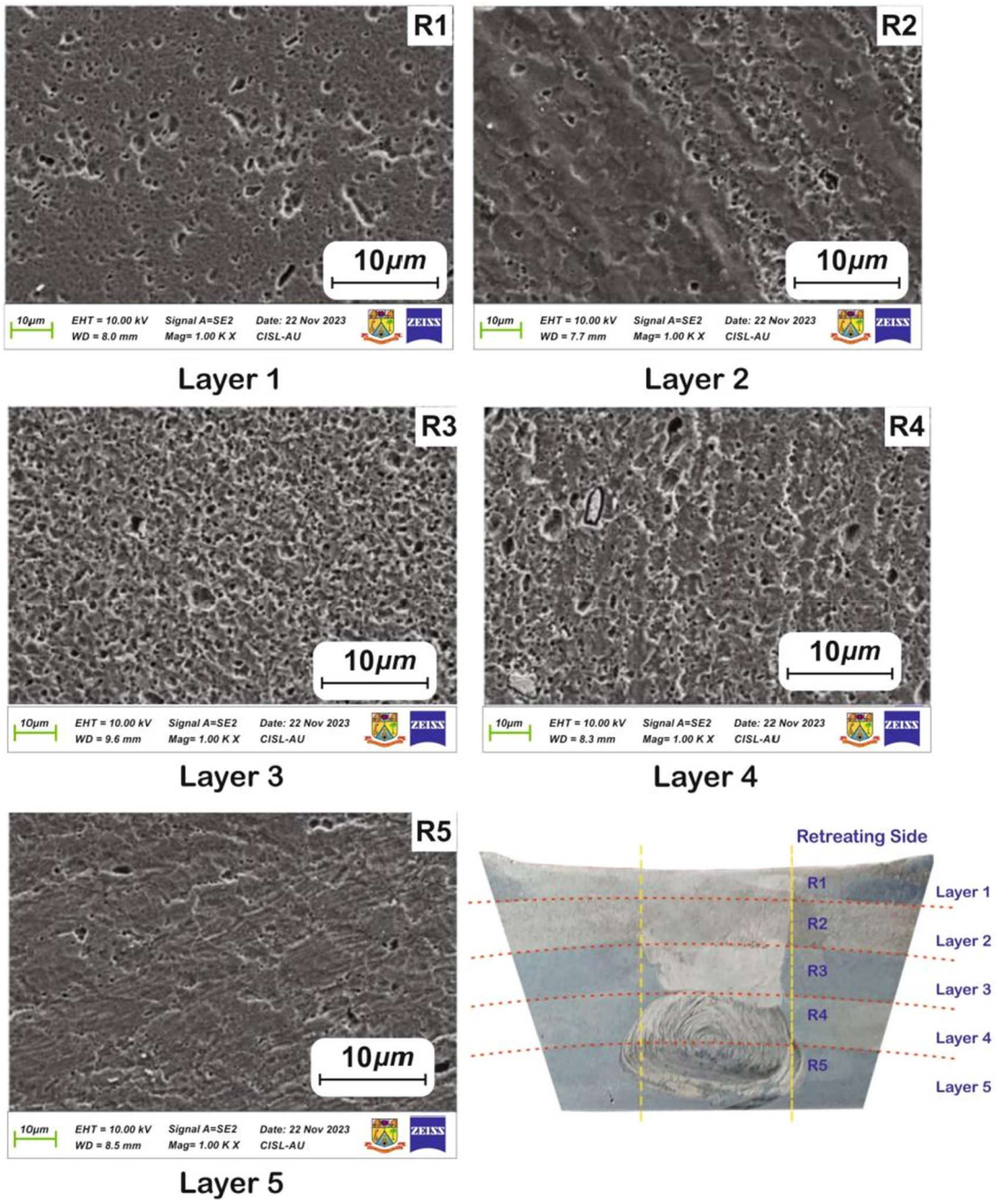
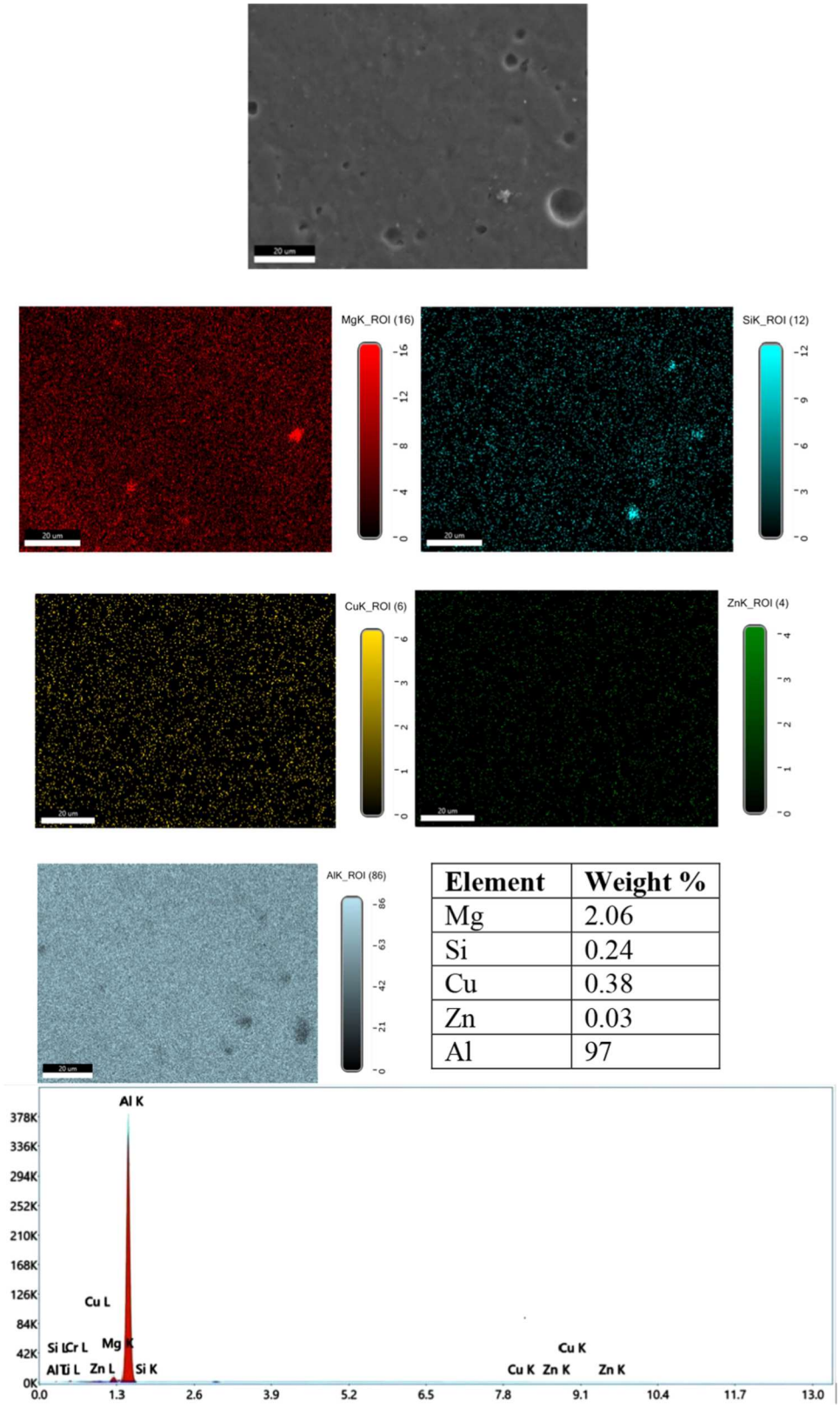


Figure 12. Layer-by-layer microstructural depiction of the RS utilizing FESEM, FSAM's aluminum-based functional gradient material.

In the stir zone of the AFGM layer-by-layer, it is obvious that the mixing is working. The surface appears really enriched: magnesium rises to a level of approximately 2.5–2.7 per cent on a base of about 1.8% (and occasionally even as low as 0.8%). Copper increases to 1% to more than 2, and zinc increases to 4.8 to higher than 5.5–6. Even silicon is boosted to 0.7%–0.8% as compared with the base alloy of 0.35%–0.6%. Deeper and deeper, and things begin to even out. In the middle layers, magnesium ranges between 2.5–3 percent, copper ranges between 1.5 to 1 percent, and zinc ranges between 3 percent to approximately 1 percent. At the fifth layer, magnesium is still suspended in there at 2%–2.5% but copper and zinc have drained out–to only 0.1%–0.2% and 0.1%, respectively. The fact that the elements are diffusion downwards is a good indication that there is a smooth compositional gradient being established. Overall, this stratified breakdown demonstrates FSAM process is not merely a combination of piles of materials but rather a process of managing the mixture of elements throughout the build. The outcome is a graded structure much more uniform than what would be the case with the raw feedstock.



Element	Weight %
Mg	2.06
Si	0.24
Cu	0.38
Zn	0.03
Al	97

Figure 13. Chemical composition for 600-rpm top layer SZ of aluminum-based functionally gradient material.

4.3.3. Retreating side

The difference in the elemental composition of the RS is rather minimal, as the pattern of material flow differs. The Mg content of the upper layer is similar to that in the AS (2.5%–2.7%); however, it is more variable in the deeper layers (1.0–4.2). The Si concentration has a similar pattern to the AS, and at the top layer, the concentration is 0.6%–0.87%, and the last layer has a concentration of 0.2%–0.4%. The contents of Cu and Zn decline in depth, but less than in the case of the AS, which can be explained by lower stirring intensity. Thus, the

RS also maintains a more reasonable share of alloying elements in its deeper layers, which leads to a relatively less intensive compositional gradient.

The elemental analysis of the RS of the AFGM demonstrates that there is a mid-deviation of the basic resources due to the lowered stirring penetration. The top layer has an increasing content of Mg from 1.8–0.8 to 2.5–2.7 with the effective diffusion. Si concentration also goes up to 0.6%-0.87% compared to the base alloy range of 0.35%–0.6% which leads to improved elemental mixing. Cu and Zn rise to 0.09–1.0 and 5.5–6, respectively, which confirms the diffusion and retention into the surface upwards. Mg of the fifth layer is 1.0–4.2, and the Si is decreasing to 0.2%–0.4% and Cu, Zn to 0.3%–0.4% and about 0.5% respectively, which is far more than that of the base material. These data indicate that the RS has a less favorable compositional gradient compared to either the SZ or AS, which are connected to a low ratio of thermal and mechanical activity; however, they validate the reality that FSAM can partially integrate elements even under unfavorable stirring

4.3.4. EDX line-scan analysis (Quantitative Diffusion Profile)

The EDX line-scan profiles at 600 RPM (figure 12 & Annexure 3) provide quantitative confirmation of the graded elemental distribution across the five-layer AFGM. Along the AS, the top layer exhibits a higher concentration of Cu (≈ 2.5wt%) and Zn (≈ 6wt%), which gradually decreases towards ≈ 1wt% Cu and ≈ 4.8wt% Zn in the bottom layers. This monotonic transition verifies the intended strength-graded architecture and indicates that heat-assisted diffusion combined with mechanical stirring promotes partial intermixing while retaining the compositional gradient within the stir zone. The line-scan reveals enrichment of Mg and Cu bearing phases in the mid-thickness region, consistent with the formation of Mg-Cu-rich precipitates that contribute to the hardness peak observed at 600RPM. In contrast, the 400RPM condition displays sharper elemental transitions and irregular fluctuations in Mg and Cu content, indicating limited diffusion and less effective interface mixing, which aligns with the lower mechanical performance at this speed.

On the retreating side, the profiles show elevated Mg (≈ 2.5–2.7wt%) and Si (≈ 0.6–0.8wt%) in the upper layers, while the deeper layers exhibit noticeable variability in Cu and Zn relative to the base material. At higher TRS (700 and 800RPM), the line-scan curves still reflect that extensive heat input destabilizes precipitates, enhances grain growth, and promotes partial segregation. This behavior is consistent with the observed coarsened microstructure and reduced strength at elevated TRS.

These Quantitative line-scan profiles clearly demonstrate that elemental diffusion in FSAM-fabricated AFGM is governed by the combined effects of the thermal cycle (peak temperature and cooling rate) and stirring intensity (material flow and strain rate). At 600RPM, the balance between heat input and mechanical mixing enables sufficient lattice and grain-boundary diffusion to form smooth and stable compositional graded. Conversely, inadequate heat at 400RPM limits diffusion, while excessive heat at 700–800RPM leads to overaging, precipitate coarsening, and local segregation. This diffusion-controlled evolution of elemental graded explains the superior hardness and tensile strength obtained at 600RPM.

4.4. Tensile properties

At different TRS in two directions—the build direction and the stir zone—the tensile characteristics of the produced AFGM were assessed. For both test orientations, figure 14 displays data for tensile strength, yield strength, and elongation. The tensile strength, yield strength, and elongation values reported here correspond to the average of three tensile specimens tested under identical conditions, and the associated standard deviations confirm the repeatability of the mechanical response. Over the build direction, the tensile strength had a standard deviation of 31 MPa and a mean of 264 MPa, and the highest value of the tensile strength at 600 rpm was 310Mpa, which was 17.3 percentage points better than the mean tensile strength. Tensile strength was also found to increase steadily (400 RPM - 242 MPa), then decreased (600 RPM -310 MPa, 11% lower than peak), then decreased (700 RPM -276 MPa, 25% lower than peak), and then decreased (800 RPM -231 MPa, 25% lower than peak). The same was correct with yield strength, which recorded a maximum of 252 MPa at 600 RPM with an overall average of 215 MPa. The values of elongation were 18 and 21, giving the best mechanical performance.

Tensile strength had a mean of 227 MPa, a maximum of 231 MPa at 600 RPM, and a minimum of 222 MPa at 800 RPM, showing a reduced range of variation when compared to build direction. The yield strength was also in the same trend with the average value of 183MPa and the highest and the lowest of 190MPa and 172MPa, respectively, at 600rpm and 800rpm. The standard deviation of 6 MPa is low, which indicates how the mechanical properties of the different speeds remain relatively steady. It was confirmed that the ductility remained consistent over the SZ since the trend was smooth and the mean was 17, with slight fluctuations of [16, 17].

During the tensile tests, the main finding was that the construction direction and the SZ had different failure behaviors. On the build direction, the specimens had partial fracture (figure 15(a)), and this implies that the material was ductile and structurally sound even under high speeds. On the other hand, all specimens that were

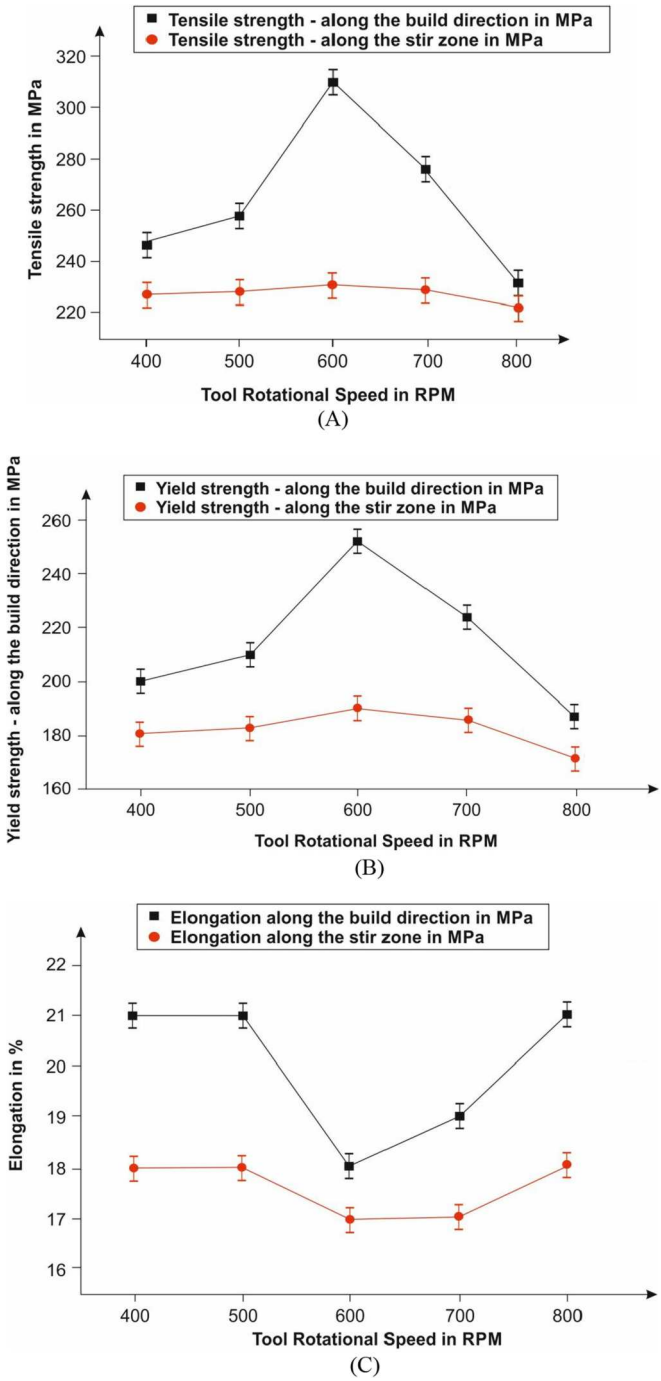
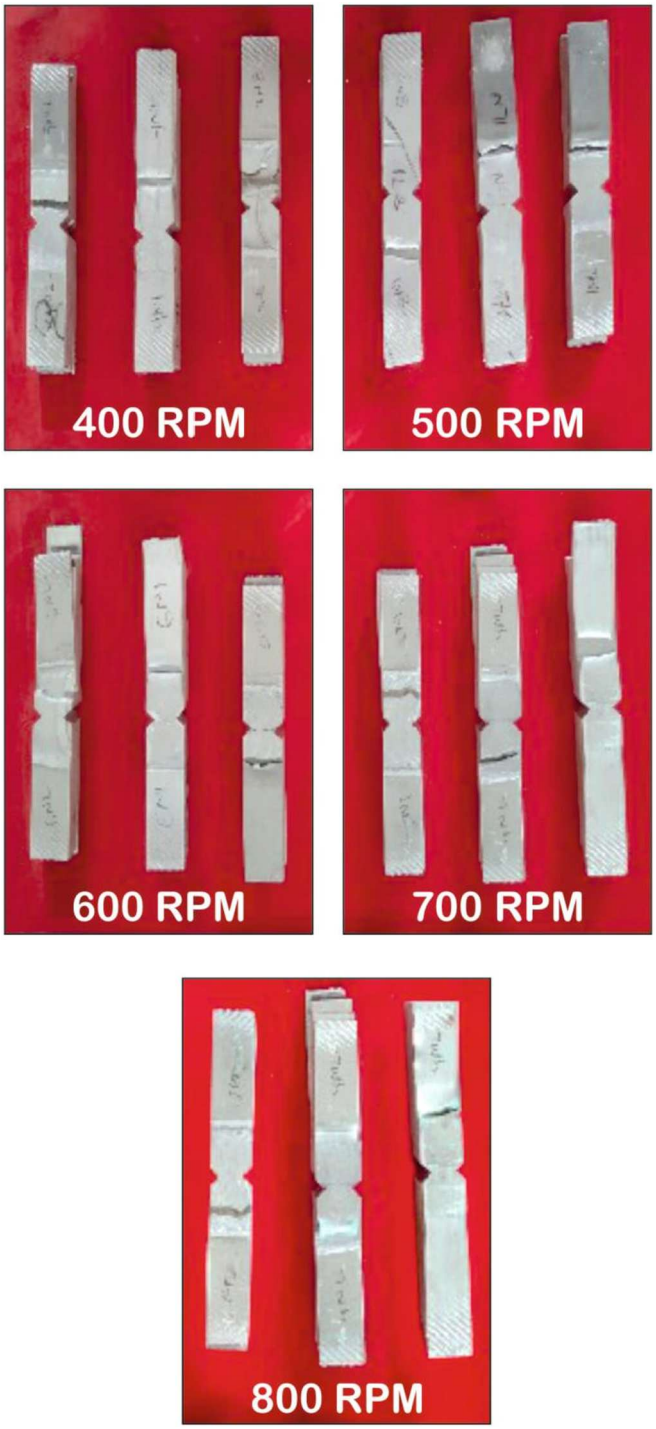


Figure 14. Tensile properties along the build direction and SZ of AFGM with different TRSs (A) tensile strength, (B) yield strength, (C) elongation.

tested in the SZ fractured completely (figure 15(b)), which is an indication of a lower bearing capacity at the local material softening in this region.

4.5. Microhardness

The microhardness assessment of the FSAM-fabricated AFGM revealed substantial variance across various TRS settings and throughout the cross-sectional thickness (figure 16). Each hardness value represents the mean of five indentations taken at the same depth and location, and the corresponding standard deviation was included to account for local variations within each layer. The highest microhardness of 139 HV was recorded at 600 RPM on the top surface, corresponding to the formation of fine and dense microstructure. Slightly lower values were obtained at other TRS, measuring 135 HV at 500 RPM, 130 HV at 700 RPM, 128 HV at 800 RPM, and a minimum of 110 HV at 400 RPM. The findings demonstrate that plastic deformation and heat input both have a substantial impact on hardness, with 600RPM being the best circumstances for surface strengthening.

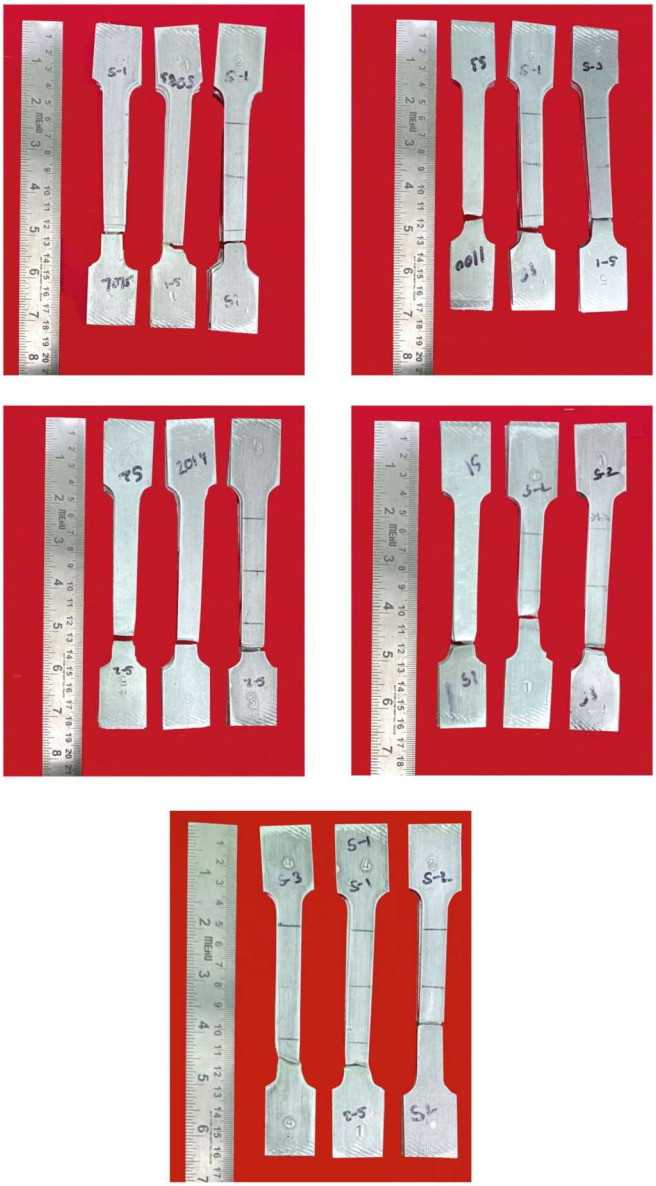


(a)Tensile specimens in the build direction

Figure 15. Post tensile specimens.

A distinct hardness gradient was observed between the layers, with a sharp decrease in the fifth layer, where hardness values dropped to 32–41 HV across all TRS conditions. This decrease implies the existence of a softer zone, as it is characterized by a lower heat contribution and reduced stirring strength at deeper levels when running through the FSAM process. The decrease in the hardness of deeper layers is also affected by the compositional design of the aluminum alloys of choice that facilitates a graded microstructure. This is consistent with the key goal of FGM design, whereby it is possible to achieve localized properties by means of variations in hardness.

The hardness region was high at 500 RPM, then a medium-hardness region was observed between 4.5 mm to 8.2 mm, and finally a low-hardness region was observed beyond 8.2 mm. This high-hardness zone was constrained at 400 RPM to 2.1 mm above the top, and it was further increased to 8.4mm, and the low-hardness



(b)Tensile specimens in the SZ

Figure 15. (Continued.)

region was further beyond this point. At a 600 RPM, the high-hardness area was 4.4 mm deep of the top, the medium-hardness area to 8.2 mm, and the low-hardness area above this edge. Equally, 700 RPM had a high region of hardness to 4.4 mm, a medium region of hardness between 4.4 mm and 8.7 mm, and a low region of hardness after that. At 800 RPM, the high-hardness region extended to 4.2 mm, followed by a medium-hardness region up to 8.6 mm, and a low-hardness region beyond this depth.

4.6. Scalability of optimal tool rotational speed

Experimental results show that 600 RPM is the best TRS for the five-layer aluminum alloy material studied, which has few flaws, a refined grain structure, and outstanding mechanical properties. Expanding this method to handle larger components or alternative alloy combinations does, however, need careful consideration of a number of issues.

In order to avoid overheating and maintain steady material flow, the TRS may need to be adjusted downward as the component size grows, since the accumulation of heat becomes more noticeable. Likewise, the ideal TRS may change when manufacturing alloys with a different hardness level or thermal conductivity compared to the ones utilized here. Reduced rotational speeds may be necessary to prevent excessive heat accumulation and guarantee appropriate material consolidation for tougher alloys or materials with lesser thermal conductivity, for example.

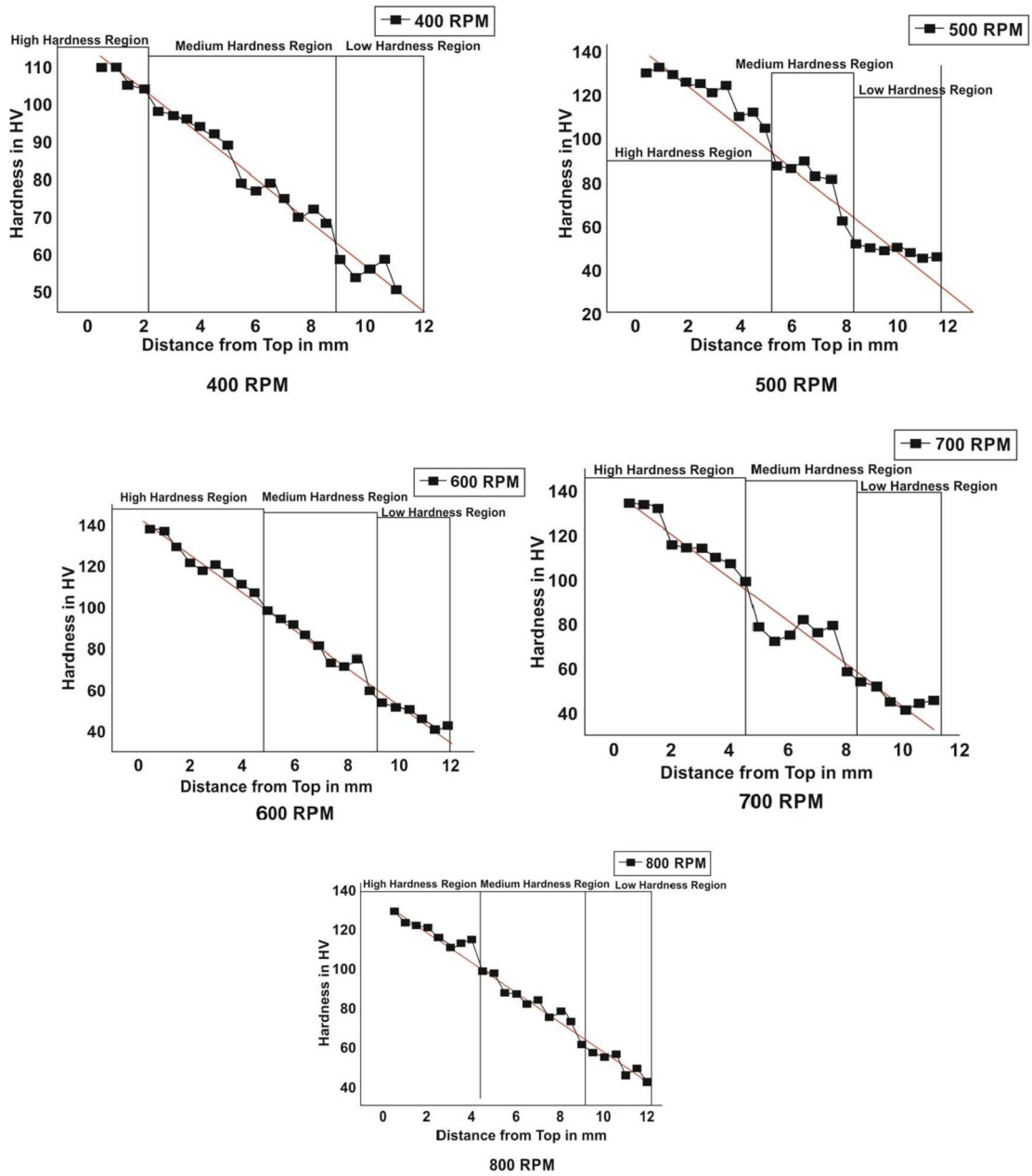


Figure 16. Microhardness versus thickness distance of top to bottom of all TRSs.

A tensile strength of 310 Mpa, a maximum possible hardness of 139 HV, and a material microstructure with no defects confirm that the 600 RPM speed provides the best balance between the amount of heat input and the direction of material flow in the present system. The results provide a strong basis for the similar multi-layer aluminum systems at 600 RPM. Process parameters should be validated and fine-tuned on an individual material and component geometry basis to guarantee the consistency of process performance in different manufacturing conditions (Geometric scaling implications must be studied thoroughly, and thermal control methods must be considered).To generalize these results to manufacturing at an industrial scale, additional research needs to be conducted that includes detailed analyses of geometric scaling implications and thermal control methods.

4.7. Benchmark against the literature

To contextualize the findings of this study, the microstructural and mechanical responses obtained at different TRS were critically benchmarked against existing FSAM literature, with a comparative summary presented in table 6. A critical evaluation of these studies demonstrated that the present work advances the state of FSAM-based aluminum FGMs beyond previously reported capabilities. Earlier investigations of a binary aluminium system typically achieved grain sizes of 4.5–9 μm and tensile strengths around 260–280MPa under optional

Table 6. A comprehensive review with related studies.

Study	Material	Technique Used	Parameters	Grain size (μm)	Hardness in HV	Tensile strength in MPa	Key findings
Our study	AFGM	FSAM	600 RPM	3–6	139	310	Fine grains, minimal defects, optimal strength
[23]	Al-6061/Al7075	FSAM	600RPM 40 mm min^{-1}	7	138	280	Refined grains & Improved strength
[24]	AA 2024/AA6061	FSAM	400–800 RPM influence of interlayer bonding on such architectures, on diffusion behavior.' 20–60 mm mm^{-1}	4.5–9.2	132	265	Lower RPM enhanced ductility, & Higher RPM caused voids.
[25]	Al-Si Composite	FSAM with Reinforcement	700 rpm 50 mm min^{-1} Threaded pin	3.8–6.0	145	295	Uniform Si desperation, improved wear resistance
[26]	Al-Cu-FGM	FSAM	800 RPM	6.1	128	270	High Cu content increased the hardness in the upper layers
[27]	AA5083/AA7075	FSAM	500–700 rpm 45 mm min^{-1}	5.2	134	275	Balanced heat input improved joint strength
[28]	AA7075/SS 1100/AA6061	FSAM	600 rpm 50 mm min^{-1}	4.8–7.3	139	288	Multi-layer bonding improved microstructural uniformity
[29]	AA2024/Al-CNT	FSAM + Nanoparticle Addition	500 rpm 25 mm min^{-1}	3.5	148	310	CNT enhanced the hardness and tensile strength significantly
[30]	Al-7075	FSAM	650 RPM	6–10	130–150	270–290	Slightly lower hardness due to coarse grains
[31]	Al-6061	FSAM	700 RPM	8–12	120–140	260–280	Phased clustering and heat included a defect
[32]	Al- AFGM	Hot Extrusion	600 RPM	4–7	145–165	285–310	Conformed uniform phase distribution
[33]	Al- Alloys (casting)	Casting	—	>15	>120	200 Retracting	Coarse grains and poor mechanical properties
[34]	Al-Si- 10-mg	Selective laser melting	—	3–6	150–170	290–320	High hardness

conditions [23, 24], while improvements observed in composite or reinforced systems were largely dependent on ceramic or CNT additions [25–29]. Multi-layer studies involving only three layers also reported tensile strength below 290MPa and exhibited microstructural inconsistencies at elevated TRS [26–28]. In contrast, the current study successfully fabricates a five-layer gradient structure comprising AA7075, AA2024, AA6061, AA5083, and AA1100 materials with widely varying thermal and metallurgical characteristics and still achieves refined grain sizes of 3–6 μm and tensile strength of ~310MPa at 600RPM without any reinforcement additions. Compared with high-TRS FSAM work that frequently reports precipitate coarsening and void formation [30, 31], the present results demonstrate uniform precipitate distribution and stable microstructural characteristics through careful TRS optimization. Although hot-extruded FGMs [32, 33] and SLM-based aluminium alloys [34] achieve comparable mechanical performance, these methods require secondary thermomechanical treatments, whereas the present approach attains similar performance within a single FSAM processing step. This comparative assessment confirms that the current study provides a more advanced and comprehensive understanding of process-structure–property relationships in a multi-alloy gradient system.

Building upon this comparative observation, a broader trend emerges regarding the influence of alloy heterogeneity, thermal graded, and material flow characteristics on FSAM performance. Studies in table 6 consistently indicate that alloy systems with narrow thermal conductivity ranges on similar precipitation kinetics trend to produce unstable heat distribution and heterogeneous stirring at elevated TRS, resulting in interfacial defects, non-uniform grain refinement, and interior strengthening response. The present multi-alloy configuration, by contrast, introduces deliberate variations in solute content, thermal conductivity, and solid-solution behavior, which effectively stabilize heat flow and strain partitioning during layer deposition. This promotes more controlled dynamic recrystallization and precipitate redistribution, enabling superior microstructural uniformity even under TRS conditions where previous studies reported void formation or over-aged intermetallic. These mechanistic differences explain why the present system sustains fine-grain structures and stable mechanical performance across a broader processing window compared with earlier FSAM-based AFGM.

From an industrial perspective, these findings demonstrate that FSAM can be strategically engineered to fabricate multi-layer aluminium structures with a predictable process-structure property relationship. Unlike many prior approaches that relied on reinforcements, complex heat treatment, or narrow TRS tolerances to maintain stability, the current process achieves comparable or superior properties through intrinsic material–thermal interactions alone. This significantly enhances manufacturability by reducing processing steps, enabling consistent interlayer bonding, and improving tolerance to thermal fluctuations—key requirements for scaling FSAM to larger components. Consequently, the present work not only surpasses prior literature benchmarks but also provides a practical and scalable pathway for producing lightweight graded panels, transition joints, and structural components for aerospace, automotive, and marine industries.

5. Discussion

An interdisciplinary microstructural investigation found that TRS influences both the precipitation behavior and grain refinement. With balanced thermal cycling, we were able to attain ideal conditions for dynamic recrystallization and homogenous grain refinement at 600 RPM in the stir zone and advancing side, where plastic deformation and heat input were highest. When the TRS value hit 800 RPM, however, it started to generate an excessive amount of heat, which cut into the cooling rates. This led to a noticeable coarsening of the grain, particularly on the retreating side, indicating an imbalance in the thermo-mechanical conditions. The mechanical properties were enhanced at an optimal speed of 600 RPM due to the tiny grains, which, according to the Hall-Petch relationship, prevent dislocation movement and increase the material's strength [35, 36]. This trend is quantitatively validated by the Hall-Petch correlation plot (figure 16), where hardness and yield strength increase systematically with $d^{-1/2}$ Confirming the direct contribution of grain refinement to the enhanced mechanical properties at 600 RPM. The average S^2 Grain size decreased from ~12–15 μm at 400RPM to 4~6 μm at 600 RPM, corresponding to a rise in yield strength from ~200 Mpa to ~225 Mpa and arise in hardness form ~110 Hv to ~139 Hv. The inverse trend was observed at 800 RPM, where excessive heat input caused grain coarsening to ~20–22 μm and resulted in reduced tensile strength (~185 Mpa) and lower hardness (~90Hv). A clear linear relationship was obtained when plotting yield strength against $d^{-1/2}$, demonstrating that the finest grains at 600 RPM occupy the highest position on the Hall-Petch plot (figure 17). Confirming that grain refinement is the dominant strengthening mechanism governing the mechanical response of FSAM-fabricated AFGM.

Various TRS values had a notable impact on the microstructural development throughout layered zones, as shown by precipitate analysis. The irregular shape and uneven distribution of precipitates were caused by a lack of heat input and material mixing at low TRS (400 RPM), especially on the retreating side and in deeper layers.

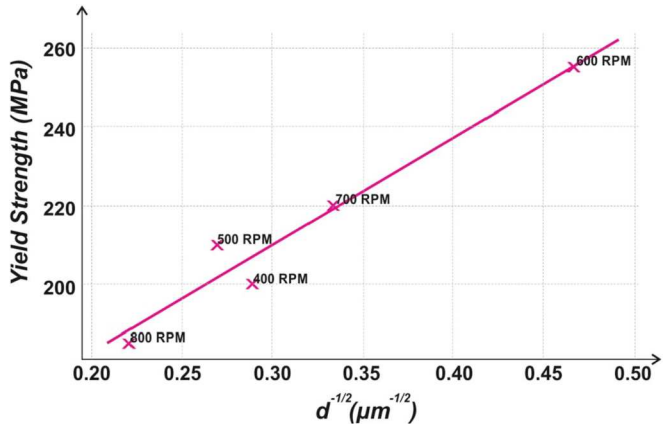


Figure 17. Yielding strength versus $d^{-1/2}$ showing Hall-Petch behaviour for FSAM-processed AFGM at different TRS.

These types of variations undermine the mechanical strength of AFGM and are characterized by the accumulation of magnesium and zinc-rich precipitates. [35, 37]. 600 RPM TRS, however, gave the most evenly distributed precipitates and the best in the SZ and AS. An intermediate that exists between the heat input and plastic deformation promoted a uniform distribution of the elements, and this resulted in enhanced structural and mechanical characteristics of the AFGM. Over-aging and precipitate coarsening (in particular on the outer in AS and RS) occurred due to excessive heat, and hence increased TRS (700 and 800 rpm) was used. The coarse uneven precipitates, which were loaded with magnesium and zinc, enabled the formation of the resulting cracking microstructure that reduced ductility and exposed crack propagation vulnerability.

The essential alloying components were spread out as precipitates in the various layers, as depicted by the EDX analysis. The heating caused the Mg, Si, Cu, and Zn elements to be deposited on the upper layer at a rate of 600 RPM. Precipitates at the intermediate were rich in magnesium and copper, and this significantly enhanced the overall strength and hardness of the material [38]. At 400 RPM, the magnesium and copper segregate less noticeably; the trend is also less regular, which reduces the strengthening effect to some extent. At a temperature of 700 or 800, heat is experienced during RPM. Their uniform dispersion is affected by precipitates as they start breaking and mixing in a few seconds. The hardness of the material is reduced, its microstructure fails, and it cannot resist the tensile tests. Thereby, 600 RPM will be the most suitable mechanical operation.

Comparing the source material to the generated AFGM, it was evident that the elemental material content of the AS, RS, and SZ was different. The upper layers of the AS were subjected to increased heat input and stronger stirring, thus increasing the content of Cu (1 to 2.5 percent) and Zn (4.8 to 6 percent). Conversely, since the materials flow was limited, these elements were much less common in the lower classes. Similar patterns were seen on the RS, where the lower layer showed noticeable variations in Cu and Zn in comparison to the base alloy, while the top layer had higher concentrations of Mg (2.5%–2.7%) and Si (0.6%–0.87%). In the SZ, intensive mixing enriched Mg, Si, Cu, and Zn in the top layers and steadily decreased them at the bottom. The higher layers were stronger than the deeper layers due to compositional graded caused by heat production and material flow.

The optimal TRS had minimal precipitate formation, but too high or too low a TRS enhanced it. It was caused by the excessive or deficient input of heat, which produces local melting. These defects were disastrous to ductility and fatigue strength, proving the need to control process parameters. The heterogeneous precipitates formed were caused by the stimulation of the local thermal inclinations, and this affected the strength of the material. Whereas under tensile loading, sharp geometries and places of precipitates usually acted as the source of crack and hence, limited ubiquitous plastic deformation [39]. Quantitative analysis showed that samples with high voids and precipitates had a 15 to 20 percent elongation when compared to the defect-free samples. This supports the intensive association of the occurrence of precipitates and the ductility loss [40].

The effects of TRS on ductility and strength were validated by tensile testing. The best results were obtained at 600 RPM, with a tensile strength of 310 MPa and a yield strength of 252 MPa. This was achieved by means of stabilizing plastic deformation, which was made possible by improved grain architectures and homogeneous element distribution. Owing to inadequate heat input, low TRS (400 RPM) resulted in coarser grains and lower tensile performance. FESEM investigation revealed that structural integrity was impaired owing to significant precipitate development and thermal softening [41].

In all conditions of TRS, the upper layers were always finer grained, and the bottom layers acquired relatively coarse grains. This size gradient of grains had a direct impact on the microhardness and tensile strength. The grains

in the upper layers were finer and therefore had higher microhardness, which reached a maximum of 139 HV at 600 RPM. The tensile strength was higher at that point, as supported by the Hall-Petch relationship. These results report the high correlation of grain refinement, grain hardness, and strength, which confirms that the change in grain size at the layers serves as the main control of the mechanical performance of the fabricated AFGM.

The post tensile results indicated cracks in the direction of the build that had not broken completely, and the layers remained intact in some of the samples, as well as fractured with the other samples having a higher rate at 600 RPM and over. After a complete fracture occurs over numerous layers with different resistances, the multilayered structure seems to take advantage of the gradient qualities[42–46]. These traits come in handy when damage tolerance and multilayer durability are of the utmost importance.

A major parameter, which regulates the equilibrium between heat, material flow, and cooling rate, is the TRS; this parameter directly affects the grain size and stability of the precaution. The optimal TRS was 600 RPM, which gave the optimal grain structure, the most consistent grain distribution, as well as the optimal tensile strength and hardness. On the other hand, the non-optimal mechanical properties and abnormal morphology were generated by high TRS (700–800 RPM) and low TRS (400–500 RPM), which indicates that the material parameter choice in FSAM requires accuracy. While the present study establishes a clear TRS microstructure property relationship for multi-alloy FSAM-fabricated AFGMs, certain limitations remain. Detailed qualification of interlayer bonding-through techniques, such as nano-indentation mapping, interfacial shear testing, or fracture surface analysis, was not performed and would strengthen understanding of interface integrity. Residual stresses, which influence fatigue life and dimensional stability in layered structures, were also not evaluated. Additionally, although thermal measurements were conducted, comprehensive thermal-mechanical modelling was not included and could provide predictive insight into heat flow, strain distribution, and precipitate stability during FSAM. Further studies should therefore incorporate multi-scale bonding assessment, residual stress analysis, and numerical process modelling. Expanding the investigation to include fatigue, corrosion, and long-term durability will support he deployment of FSAM-based AFGM in practical applications.

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Future work will focus on advanced inter layer bonding assessment, residual-stress evaluation and thermal mechanical modelling to further optimize the FSAM process for large scale structural applications.

6. Conclusion

In this study, researchers looked at functionally graded materials made of aluminum using FSAM and how the mechanical performance and microstructural development are affected by the tool rotational speed. Here are the main results:

(i) For all layers, an optimal TRS of 600 RPM resulted in consistently refined grain shapes and homogenous distributions of precipitates.

(ii) FESEM examination verified that a speed of 600 RPM reduced the occurrence of defects. Low TRS values resulted in inadequate consolidation, while high TRS values reduced ductility and tensile strength by inducing heat deterioration and microcracking.

(iii) The results of the EDX study showed that the elemental gradients were regulated, and that the distribution of hardness and the processes for strengthening were affected by the enrichment of magnesium in the intermediate layers.

(iv) At 600 RPM, the microhardness reached its maximum of 139 HV. Lower TRS values caused coarser grains and reduced hardness, in contrast to higher TRS values that caused thermal softening in deeper layers.

(v) At 600 RPM, the mechanical parameters were optimized, with a tensile strength of 310 MPa, a yield strength of 252 MPa, and an elongation of 18%. Increasing or decreasing the TRS value has the same effect on strength and ductility.

Overall, when it comes to manufacturing high-performance aluminum FGMs for industrial uses, the ideal processing parameter is FSAM at 600 RPM since it minimizes defects while simultaneously increasing ductility and strength. To further enhance FSAM procedures for large-scale structural components, future studies should explore interlayer bonding features, residual stress distributions, and thermo-mechanical modeling.

Conflict of interest

The authors declare no conflicts of interest regarding this manuscript.

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Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

Author contributions

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